

Nabheet Singh Sandhu

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Summary

AI Engineer building production agentic systems and developer tooling that bridge AI with real-world environments. Creator of open-source MCP servers connecting AI agents to IDEs, manufacturing APIs, and automation pipelines. Combines systems engineering experience with modern AI/ML to deliver cloud-native intelligent systems at scale.

Skills

AI/ML: LangChain, OpenAI API, Prompt Engineering, RAG, Agent Frameworks, MCP, Google ADK
Languages: Python, TypeScript, JavaScript, Ruby
Frameworks: Node.js, FastAPI, React
Cloud Platforms: GCP (Cloud Run, Pub/Sub, Cloud SQL, Cloud Armor), AWS (EKS, EC2, Lambda, RDS, WAF), Pulumi
Infrastructure: Kubernetes, Docker, Ansible, CI/CD, OpenTelemetry, Grafana, ELK Stack
Databases: PostgreSQL (pgvector), MySQL, Redis

Experience

Software Engineer IV (Contract)	Spectrum	Mar 2026–Present
- Built SpectrumGPT—internal AI platform for business process automation using LLMs and agentic workflows, serving 75k+ daily users with real-time data integration. - Built AI-powered development environment with specialized coding agents for automated code review, cross-browser testing with visual comparison, and real-time production monitoring. - Improved platform reliability across the stack: eliminated intermittent server crashes through database driver upgrades, reduced API response times by ~25% with intelligent caching and early request validation, eliminated silent service failures with automatic connection recovery and health monitoring. - Modernized full-stack application upgrading 8 major framework dependencies (Node.js, TypeScript, React, Vite, Storybook, MSW, Vitest, Sass), fixing 56 type safety errors while maintaining 321 passing tests. - Delivered 14+ feature tickets across Node.js, Python, and React covering feature development, reliability engineering, performance optimization, and security hardening.		
Sr Software Engineer (Contract)	CoBank	Dec 2025–Mar 2026
- Founding member of new development team responsible for setup and definition of project architecture and development workflows. - Implemented lending product workflows migrating on-prem systems to AWS. - Utilized GitHub Actions for CI/CD and Terraform for infrastructure as code. - Improved onboarding process for new engineers by creating documentation and walkthroughs for development and deployment processes.		
Sr Software Engineer	Functional Finance	Sep 2023–Oct 2025
- Reduced release cycle testing time by implementing automated QA test generation with AI agents. - Architected OpenTelemetry observability layer providing distributed tracing and metrics across the full stack, enabling resolution of performance bottlenecks and deadlock issues. - Designed and deployed Python (FastAPI) and Node.js APIs on GCP Cloud Run, Cloud SQL, Pub/Sub, and Cloud Armor. - Built VS Code Dev Containers-based development environment, reducing engineer onboarding time and ensuring consistent tooling across the team. - Optimized FastAPI container startup in Cloud Run by 50% through dependency and configuration improvements. - Implemented PDF generation service using Node.js, Puppeteer, and Chromium on GCP Cloud Run.		

- Resolved systemic HTTPX client exhaustion bug causing source port depletion in production.
- Led a team of 2 engineers to deliver key business objectives.

Software Engineer V/IV

GoDaddy

2015–2023

- Architected CloudLinux 8 shared hosting platform, migrating 1.5M+ customer accounts from legacy infrastructure.
- Reduced customer website page load times by 50% through Apache, MySQL, and kernel-level optimization.
- Built ML-based server performance models using PCA (dimensionality reduction) and linear regression to analyze response time vs. server load relationships.
- Developed Jupyter notebook analyzing TCP connect times across global datacenters, disproving distance-based latency assumptions.
- Discovered and reported upstream Apache HTTP/2 memory growth bug (mod_h2 issue #212) and htaccess-induced memory exhaustion issue.
- Architected Ansible AWX on AWS EKS, reducing shared hosting environment patching cycles from 30+ days to under 7 days.
- Designed WAF/CDN 2.0 product architecture on AWS and deployed on-prem Kubernetes on OpenStack for provisioning systems.
- Presented on log aggregation and analysis at GoDaddy TechFest; mentored junior engineers across teams.

Earlier Career

Sr. Software Engineer / Architect

FSW, GoDaddy, Visa, Qwest

2003–2014

- Architected enterprise ELK stack processing 30GB/day for real-time log aggregation, visualization, and application performance monitoring across distributed systems.
- Built high-availability payment processing systems at Visa (provisional patent filed), large-scale telecom billing at Qwest, and e-commerce platforms at GoDaddy.
- Provided technical leadership and coaching on engineering best practices across organizations.

Open Source & Side Projects

- **vscode-mcp-server** – VS Code extension exposing 42 IDE capabilities (debugger control, LSP features, terminal execution, file operations) as MCP tools for AI agents. (TypeScript, VS Code API)
- **pcbway-mcp-server** – MCP server covering all 16 PCBWay Partner API endpoints for PCB manufacturing quotes, SMT assembly pricing, and full order lifecycle management. (TypeScript, MCP)
- **Wiki On This Day** – Agentic LangChain pipeline that fetches Wikipedia events, summarizes them via LLM, and generates documentary videos using OpenAI Sora. (Python, LangChain, OpenAI API)

Education

University of California, Berkeley

B.S. Electrical Engineering & Computer Science

Certifications: AI Engineer Bootcamp, AI Agent Engineer Course, Machine Learning (Coursera)

Accomplishments

- Selected for Tiger team to improve shared hosting platform performance at GoDaddy.
- Provisional patent filed by Visa for payment processing innovation.
- Dean's List: 8 semesters, UC Berkeley.
- Fifth engineer to join startup at Functional Finance.